

Understanding & Designing the Joke

Joseph Clay · June 22, 2026 · 659 words

1 · Systems researched

JAPE

Purpose. Generates jokes — specifically punning question-and-answer riddles.

Mechanism. Hand-built linguistic schemas over a structured lexicon. Templates such as "What do you call a X? A Y" are filled by matching words on phonetic and semantic features to engineer a pun.

Strengths. Produces genuinely novel, grammatical puns and is grounded in a formal linguistic theory of humour.

Limitations. Restricted to one joke form; rigid templates; needs richly annotated lexical data; humour quality varies widely.

GPT-3

Purpose. A general language model that can produce jokes via prompting, though not built specifically for humour.

Mechanism. A 175-billion-parameter transformer trained to predict the next token over massive text; humour emerges from few-shot prompting rather than any humour-specific design.

Strengths. Highly flexible across joke types; fluent and fast; needs no templates.

Limitations. Inconsistent quality; no real grasp of why something is funny; recycles clichés; can be incoherent or offensive.

Bard / Gemini

Purpose. An instruction-tuned chat model that can both generate jokes and explain humour conversationally.

Mechanism. A large transformer (PaLM 2 / Gemini) tuned with instruction-following and human feedback (RLHF) for dialogue.

Strengths. Conversational; can explain the mechanism of a joke; can draw on recent information.

Limitations. Shares LLM caveats — often surface-level humour, heavy safety filtering, variable originality.

2 · Manual joke algorithm

1. **Start:** Start: pick an everyday scenario - Working as a banker
2. **Decision:** Does it set a clear expectation? — No → Make the setup more specific: name the exact job
3. **Decision:** Is there an unexpected twist? — No → Add an incongruity: Borrow a finance term
4. **Action:** Apply technique: category shift "interest" = job term + feeling
5. **End:** End: deliver setup + punchline "I used to be a banker, but I lost interest."
6. **Technique:** Incongruity
7. **Setup & punchline:** as below.

I used to be a banker

But I lost interest

3 • Audience evaluation

Average rating: **3.7 / 5** across 3 respondent(s).

Sarah W. (32) — 4/5. She laughed at the joke because she use to work at a bank understands the financial term of interest, but also how sometimes banks can be boring and it causes you to lose interest.

Jason T. (29) — 4/5. Jason found this joke to be very interesting. He is a day trader and could instantly make the comparison between working at a bank and the financial terminology of interest and then losing interest because banks are so boring.

David C (40) — 3/5. David appreciated the joke and felt that it made complete sense in financial terms to lose interest, but also that when you are losing interest, you are losing interest in something that could captivate your attention.

4 • Reflection

Designing this joke algorithm taught me that humor has a mechanical skeleton underneath what feels like a purely intuitive art. Breaking the process into discrete steps — pick a scenario, confirm the setup creates a clear expectation, locate a word that legitimately supports two readings, then resolve with the unexpected one — showed that a joke isn't really one idea, it's two valid interpretations collapsing into the same sentence at the same time. The algorithm didn't write the joke for me, but it gave me a repeatable test for whether a joke would work before I ever said it out loud.

The human feedback confirmed both the strength and the limit of that structure. Sarah and Jason, who each have direct exposure to banking and financial terminology, immediately recognized "interest" as doing double duty and rated the joke highly — their familiarity with the literal meaning made the emotional pivot land faster. David also understood the wordplay but rated it slightly lower, and his comment revealed something the algorithm can't account for: he started reflecting on the metaphor itself, on what it actually means to lose interest in something that should hold your attention. That's a respondent finding meaning beyond the joke's mechanics, which a rule-based system has no way to anticipate or measure.

That's the real gap for machines here. An algorithm can verify that a setup is specific and that a word has dual meanings — that's pattern-matching. What it can't do is sense whether an audience's life experience will make the connection feel obvious or forced, and it can't predict when a joke will accidentally say something true. Humor's logic can be mapped, but its impact still depends on the listener, not just the words.